

PRIYANSHU VATS

Product & Operations Builder · AI/Robotics Field Operations · 0→1 Internal Tools

Bengaluru, KA · [+91-6206624775](tel:+91-6206624775) · priyanshu.vats03@gmail.com · [linkedin.com/in/priyanshuvats-5a68aa292](https://www.linkedin.com/in/priyanshuvats-5a68aa292) · github.com/Pvats2003

ECE graduate running egocentric AI/robotics data-collection operations across Andhra Pradesh and Karnataka: 50+ field staff coordinated and 10,000+ recording hours logged. Diagnosed fragmented WhatsApp and Sheets workflows, prototyped tools to fix them, and consolidated the lessons into **City Ops OS**, whose design is now being integrated into Instawork's internal ops tool. Ships products end-to-end through AI-assisted development (Claude Code).

EXPERIENCE

Operations Management Intern, Field Operations | Instawork Robotics Labs · May 2026 – Present

Egocentric vision-language-action (VLA) training data for robotics · Andhra Pradesh & Karnataka

- Run day-to-day field execution for robot-training data collection across Andhra Pradesh cities, primarily Rajampet and Kadapa, coordinating **50+ Field Officers, Data Captains, and Data Collectors** and logging **10,000+ recording hours**; own task/activity libraries and SOD/MOD/EOD reporting.
- Sourced and validated walk-in, fixed-premises recording sites: authored the South India VLA Field Recording Guide (**116 businesses, 19 categories**), cluster-wise lead directories with two-shift daily field plans for the Rajampet–Kadapa corridor, and high-dexterity business directories (tailors, cobblers, goldsmiths, repair shops) across **10 Bengaluru neighbourhoods**.
- Diagnosed that field updates lived in unstructured WhatsApp threads and scattered Sheets; prototyped a live command center to centralise tracking: a 10-sheet workbook, an Apps Script JSON API with automated Slack/email alerts, and a dashboard refreshing every 30 seconds.
- Prototyped a React + Claude Vision OCR form that reads recording metadata from phone screenshots, and OpsIntel, a WhatsApp parser with a Node.js/WebSocket dashboard. Testing all three showed that one city's ops would sprawl across four separate systems, which led directly to City Ops OS.

Electrical / Powerhouse Intern | ITC Limited, Munger, Bihar · Jun 2024 – Jul 2024

- Mapped power distribution across 15+ sub-distribution boards (RUPS, bypass, DG supply); analysed switchover logic and single points of failure, documenting findings for operator training and stakeholder presentations.

PRODUCTS

City Ops OS — Local-First Operating System for City Field Ops | Live · 2026 – Present

React · Zustand · localStorage · Firebase Hosting · city-ops-cf81f.web.app

- Recognised that one city's operations were spread across Sheets, Apps Script, a standalone OCR form, and a Node.js backend; consolidated them into a single per-city operating view. **Design now being integrated into Instawork's internal ops tool.**
- Chose a local-first, zero-cost architecture (all state in the browser, no backend, servers, or paid APIs) so the tool needs no accounts, infra budget, or setup; deliberately traded away cross-device sync.

Career OS — Autonomous Job Discovery & Application Agent · Ongoing

- Designed a modular system spanning multi-source job discovery with cross-source dedup, deterministic + AI-assisted matching, resume/cover-letter generation, an application assistant, and a web dashboard.
- Enforced zero fabrication: every candidate fact carries source, confidence, and verified metadata; hard stops (salary, visa, CAPTCHA/MFA) escalate to a human and nothing auto-submits. Validated by **350+ passing automated tests**.

Audit AI Copilot — AI Document Review Tool | FastAPI · Next.js · Groq · Ongoing

- Directed the build of a copilot where auditors upload documents and query anomalies in natural language, pairing a deterministic rule engine with an LLM review layer and PDF reports; scoped across three personas with MoSCoW, a full PRD, and a Y Combinator application.

Karavali — Community-Driven Local Discovery App | Solo founder · Apr 2026 – Present

Live MVP · karaval.base44.app

- Identified a gap in community-sourced discovery for coastal Karnataka; pivoted from a campus travel app to a discovery platform with AI recommendations and gamification for Udupi and Dakshina Kannada; **shipped a live MVP in one week**.

ANPR — Automatic Number Plate Recognition | B.Tech Final Year Project · Jan 2026 – May 2026

- Built a real-time licence plate detection and reading pipeline with YOLOv8n (**89.1% mAP@50**), OpenCV, and Tesseract OCR.

SKILLS

Product: Product Discovery, User Research, Personas, UX Flows, Wireframing, PRDs, MoSCoW Prioritisation, Roadmapping, KPI Definition, Agile/Scrum, Sprint Planning

Operations: Field Operations, Workforce Coordination, Site Sourcing, Daily Ops Reporting, Process Documentation, Stakeholder Management

AI/Product Development: Claude Code, LLM/OCR integrations, React web apps, Apps Script automations

Tools: Google Sheets, Excel, Slack, Figma, Canva

EDUCATION & LEADERSHIP

B.Tech, Electronics & Communication Engineering | MIT Manipal · May 2026

- Certifications: Google UX Design (Coursera), IBM Business Analyst, IBM Program Manager.
- Revels, MIT Manipal's cultural fest (Painting & Publicity): led a 10-member core team (2025) and 30 volunteers (2024).